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Conditional probability requires P(B)>0.
P(A|B)=P(A intersection B)/P(B).
If P(B)=0, this elementary conditional is undefined.
On two fair independent coins: P(A)=P(B)=1/2.
For first-heads A and second-heads B: P(A intersection B)=1/4.

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Bayes v1: P(D)=0.02; P(+|D)=0.90; P(+|not-D)=0.10.

D and not-D partition the population.

P(+)=0.018+0.098=0.116>0.

P(D|+)=0.018/0.116=9/58.

Version 2 will change false-positive rate to 0.05; posterior 18/67.